

Lecture 1

1. Problems Solved Using Discrete Mathematics

Discrete Probability: e.g. probability of winning a lottery

Graph: Is there a link between two computers in a network?

Algorithms and Complexity: how can a list of integers be sorted so that the integers are in increasing order?

2. Problem I

Prove that "For an integer n , if $3n+2$ is odd, then n is odd".

(direct proof)
Proof I: Note that $3n+3$ is even, and so is $n+1$. so n is odd.

Proof II: It is equivalent to prove that "If n is even, then $3n+2$ is even." Suppose that $n=2k$ for some int k , then $3n+2 = 6k+2$, which is even *(proof by contrapositive)*

Proof III: Assume to the contrary that there exists an integer n such that $3n+2$ is odd and n is even. $\Rightarrow n$ is even, so is $3n$. 2 must be odd, leading to a contradiction *(~ by contradiction)*

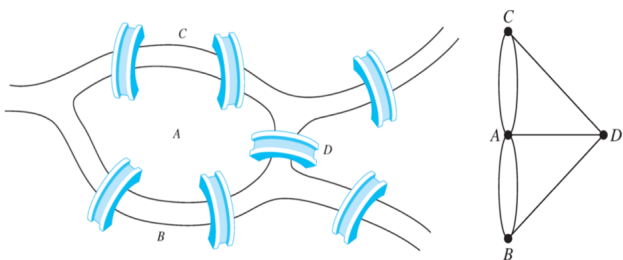
3. Problem II: Four-Color Theorem

No more than 4 colours are required to colour the regions of the map so that no two adjacent regions have the same colour.

4. Problem III: Königsberg seven-bridge problem

Is it possible to start at some location in the town, travel across all the bridges once without crossing any bridge twice, and return to the starting point.

such a solution exists if and only if its vertices has even degree.



5. Problem IV: Fibonacci Number - Population of Rabbits
 $\Rightarrow F_0 = 1, F_1 = 1, F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$

6. Propositional: a declarative sentence that is either true or false.

* Does not ask a question or give an order

Truth value: T / F

Propositional variables: e.g. p, q, r, s

Examples of propositions:

$2+2=3$

It is raining today

* 含自由变量 x , 在 x 没有被赋值前, 它可能真也可能假, 不是命题.

Not propositions:

No parking.

How old are you?

$x+2=5$

Computer x is functioning properly.
This statement is false (paradox!)

7. Compound Propositions: are built using logical connectives:

• Negation $\neg$

• Conjunction $\wedge$

• Disjunction $\vee$

$p \wedge q$ true while both p, q is true

$p \vee q$ is false when both p and q are false.

• exclusive or $\oplus$

• Implication $\rightarrow$

• Biconditional $\leftrightarrow$

$p \oplus q$ true when exactly one of p, q is true

Lecture 2

1. Conditional statement $p \rightarrow q$: "if p , then q "
this is false when p is true and q is false, and true otherwise
hypothesis conclusion

* "If" indicates sufficient condition, "only if" indicates necessary condition

The converse of $p \rightarrow q$ is $q \rightarrow p$

The contrapositive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$ ^{$p \rightarrow q$} equivalent to

The inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$.

* Equivalent: have the same truth table

2. Biconditional

$p \leftrightarrow q$: "p if and only if q", it is true when p and q

have the same truth table, and false otherwise

$\Rightarrow$ p is necessary and sufficient for q

• if p then q, and conversely

• p iff* q * = if and only iff

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

3. (1) Tautology: a compound proposition that is always true

(2) Contradiction: ~ always false

(3) Contingency: neither a tautology nor a contradiction

$\Rightarrow$ p and q are called logically equivalent, denoted by

$p \equiv q$ or $p \leftrightarrow q$. if $p \leftrightarrow q$ is a tautology

4. (1) Distributive Laws

• $w \wedge (u \vee v)$ is equivalent to $(w \wedge u) \vee (w \wedge v)$

$w \vee (u \wedge v)$ is equivalent to $(w \vee u) \wedge (w \vee v)$

(2) De Morgan's Laws:

$$\neg (p \vee q) \equiv \neg p \wedge \neg q$$

$$\neg (p \wedge q) \equiv \neg p \vee \neg q$$

(3) Important Logical Equivalences

Identity laws $\begin{cases} p \wedge T \equiv p \\ p \vee F \equiv p \end{cases}$

Domination laws $\begin{cases} p \vee T \equiv T \\ p \wedge F \equiv F \end{cases}$

Idempotent laws $\begin{cases} p \vee p \equiv p \\ p \wedge p \equiv p \end{cases}$

Double negation laws : $\neg(\neg p) \equiv p$

Commutative laws: $\begin{cases} p \vee q \equiv q \vee p \\ p \wedge q \equiv q \wedge p \end{cases}$

Associative laws $\begin{cases} (p \vee q) \vee r \equiv p \vee (q \vee r) \\ (p \wedge q) \wedge r \equiv p \wedge (q \wedge r) \end{cases}$

Others $p \rightarrow q \equiv \neg p \vee q$

5. Limitations : cannot express the values of a wider range of logical statements

$\Rightarrow$ Predicates (谓词) $\Rightarrow$ 能表达“一个对象是否具有某性质”

Quantifiers (量词) $\Rightarrow$ 能表达“所有对象/存在对象”具备某性质

Predicate Logic: make statements with variables

- A **predicate** is a statement $P(x_1, x_2, \dots, x_n)$ that contains n variables $x_1, x_2, \dots, x_n$. It becomes a proposition when specific values are substituted for the variables $x_1, x_2, \dots, x_n$.

- The **domain (universe)** D of the predicate variables $x_1, x_2, \dots, x_n$ is the set of all values that may be substituted in place of the variables.
- The **truth set** of $P(x_1, x_2, \dots, x_n)$ is the set of all values of the predicate variables $(x_1, x_2, \dots, x_n)$ such that the proposition $P(x_1, x_2, \dots, x_n)$ is true.

ex. Let $P(x)$ be the predicate " $x^2 > x$ " with domain of the real numbers.

- 1 What are the truth values of $P(2)$ and $P(1)$?

$$P(2) = \text{T}, P(1) = \text{F}$$

- 2 What is the truth set of $P(x)$?

$$x > 1 \text{ or } x < 0$$